

Dear Editors,

We are pleased to submit our manuscript, "**Scaling behaviors in simulated collagen fibrils**," for consideration as a research article in Physical Review E.

Our work investigates the fundamental relationship between the assembly kinetics of collagen fibrils and their resulting mechanical failure dynamics using a combination of modified diffusion-limited aggregation (DLA) and probabilistic fracture modeling. In this study, we highlight several significant achievements that bridge soft matter physics and biomechanics, including morphological transition control where we demonstrate that a single kinetic parameter—post-attachment surface diffusion (T_s)—governs a morphological transition from porous, fractal aggregates ($D_f \approx 1.71$) to compact, nearly Euclidean structures ($D_f \approx 1.96$). Furthermore, we identify scale-free failure dynamics showing that collagen fibril rupture evolves through stochastic avalanche events, a hallmark of Self-Organized Criticality (SOC), with avalanche size distributions following power-law scaling.

A key finding of our work is the universality crossover, where the scaling exponent (γ) is sensitive to the fibril's architecture; by increasing T_s , the system crosses the mean-field value ($\gamma = 5/2$), signaling a fundamental transition from local to global load-sharing universality classes. Our model provides a quantitative structural-mechanical bridge between the fractal dimension of a protein assembly and its mechanical resilience, offering a physical mechanism to explain how biological tissues might tune their robustness through assembly conditions.

Given the focus on statistical mechanics, fractal growth, and the physics of disordered systems, we believe our manuscript is highly suitable for the audience of Physical Review E. Our findings on the interplay between growth kinetics and universality in fracture processes address core questions in the physics of complex, hierarchical biological materials.

All authors have contributed equally to this work and have approved the final version for submission; we confirm that this manuscript is original and is not under consideration by any other journal.

Thank you for your time and consideration of our work.

Sincerely,

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